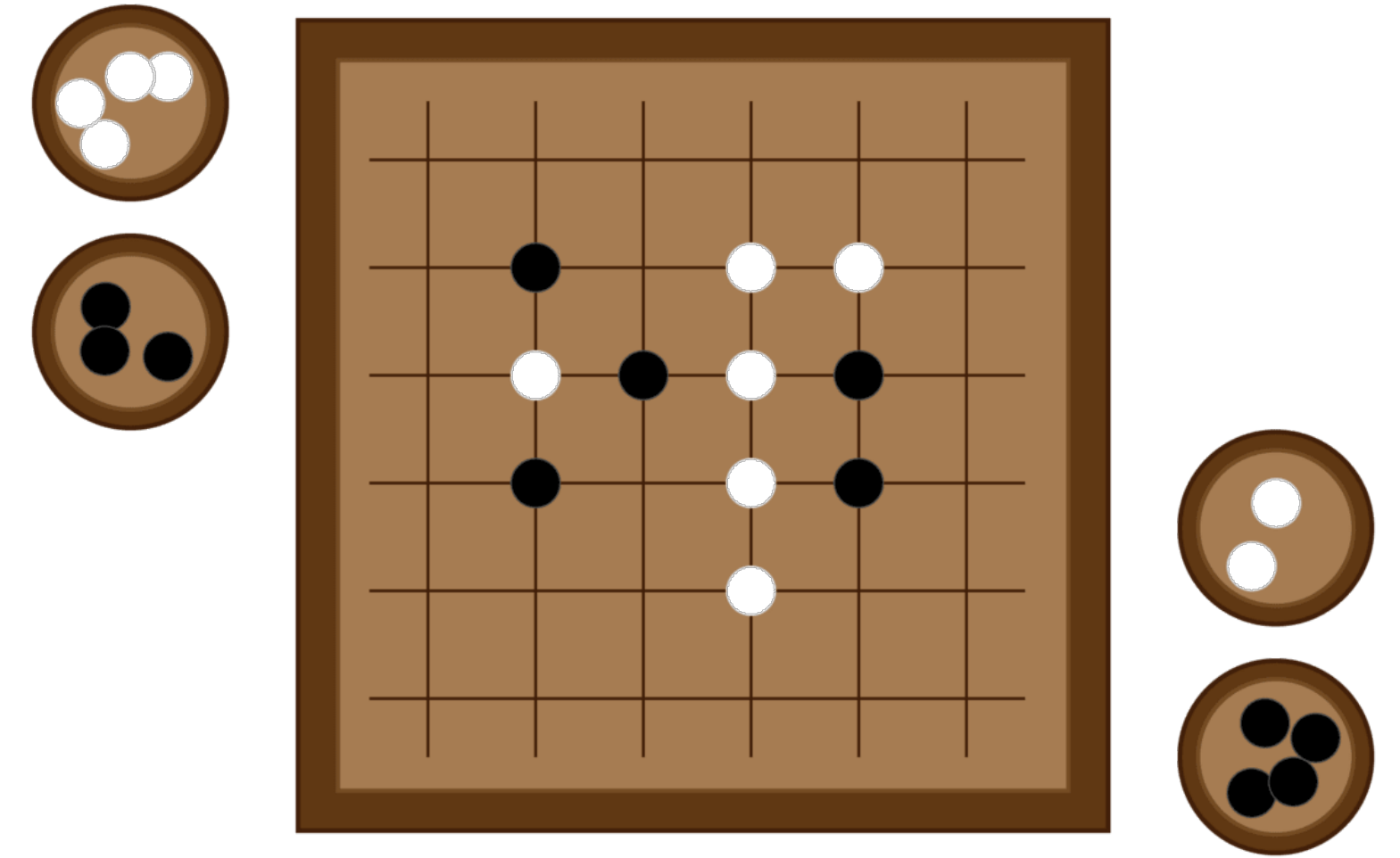




Mastering the game of Go without Human Knowledge

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Presented by: Alessia Crimaldi, Davide Luccioli

AlphaGo Zero: learning from first principles

- Tabula rasa
- Based solely on Reinforcement Learning
- Learns to play without:
 - Human knowledge
 - Human guidance
 - Domain knowledge beyond game rules



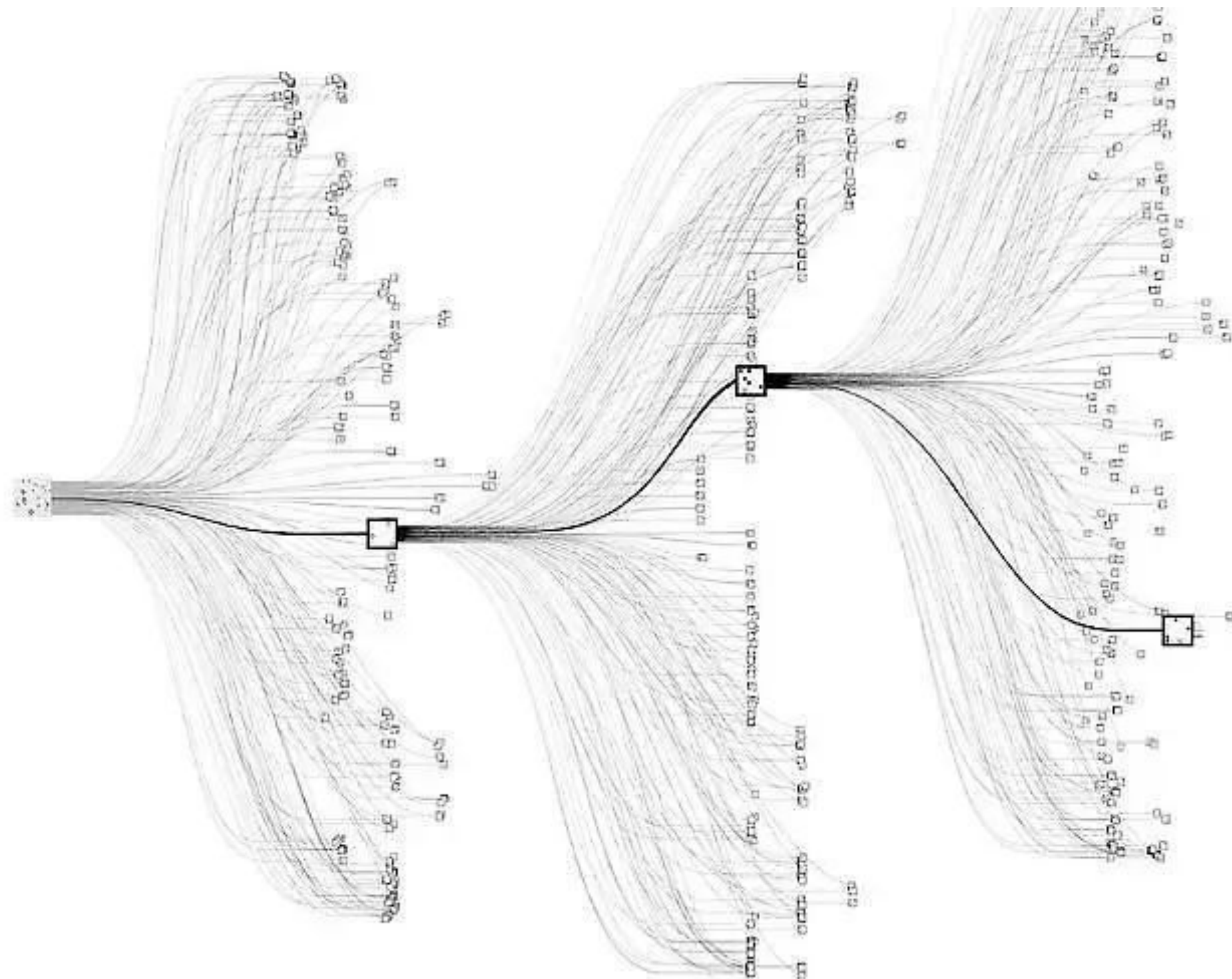
AlphaGo Zero

Motivation 1

- The reason why AI researchers care about the game of Go is that it is incredibly complex and it is hard for computers
- **LARGE BOARD:**
the number of possible configurations on the board is more than the number of atoms in the universe

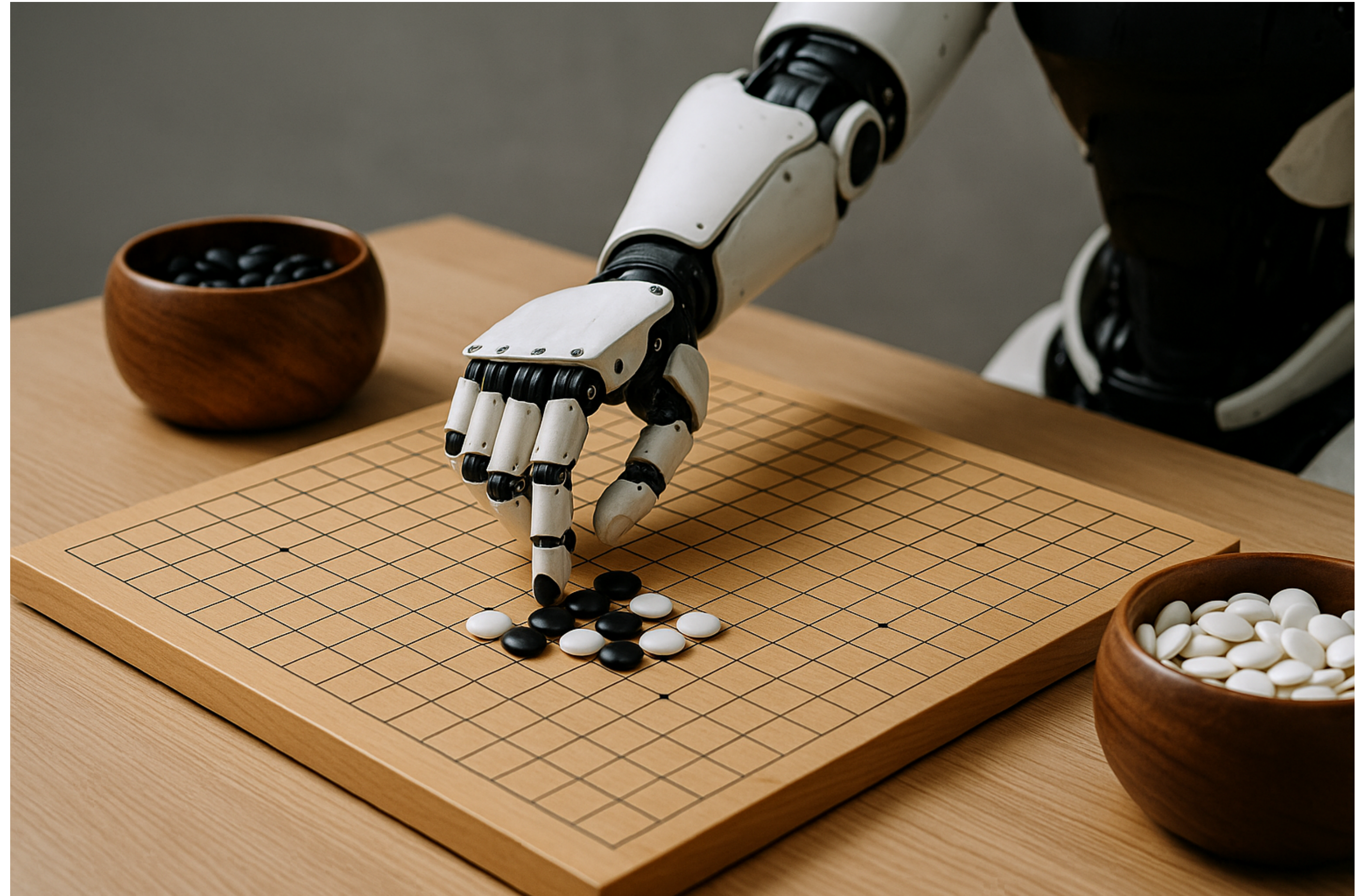
$$19 \times 19 \text{ board} \rightarrow 3^{19 \times 19} \approx 10^{170}$$

- **LARGE BRANCHING FACTOR:**
there are about 150-250 moves on average playable from a given game state
- Such a challenging domain in terms of human intellect demands precise and sophisticated lookahead across a vast search space



Motivation 2

- We use neural networks trained by **reinforcement learning** because:
 - Human data for training is often *expensive, unreliable*, or simply *unavailable*
 - Even when human data exists, it can limit the performance of the systems trained on it
- Reinforcement learning allows systems to **learn from their own experience**, enabling them to:
 - Surpass human capabilities
 - Perform in domains where human expertise is lacking



Comparison with AlphaGo Fan

AlphaGo Fan

- First algorithm to achieve superhuman performance in Go
- **Training:** Supervised learning + self-play
- **Model:** Policy and value networks
- **Input:** Hand-crafted features
- **Rollouts** are used, lookahead search provided from the trained network combined with Monte Carlo Tree Search (MCTS)

AlphaGo Zero

- Improves upon the original architecture of AlphaGo Fan
- **Training:** Entirely self-play, starting from random play
- **Model:** Single Neural Network for both policy and value
- **Input:** Raw game state representation
- **No rollouts**, lookahead search incorporated in the RL algorithm inside the training loop

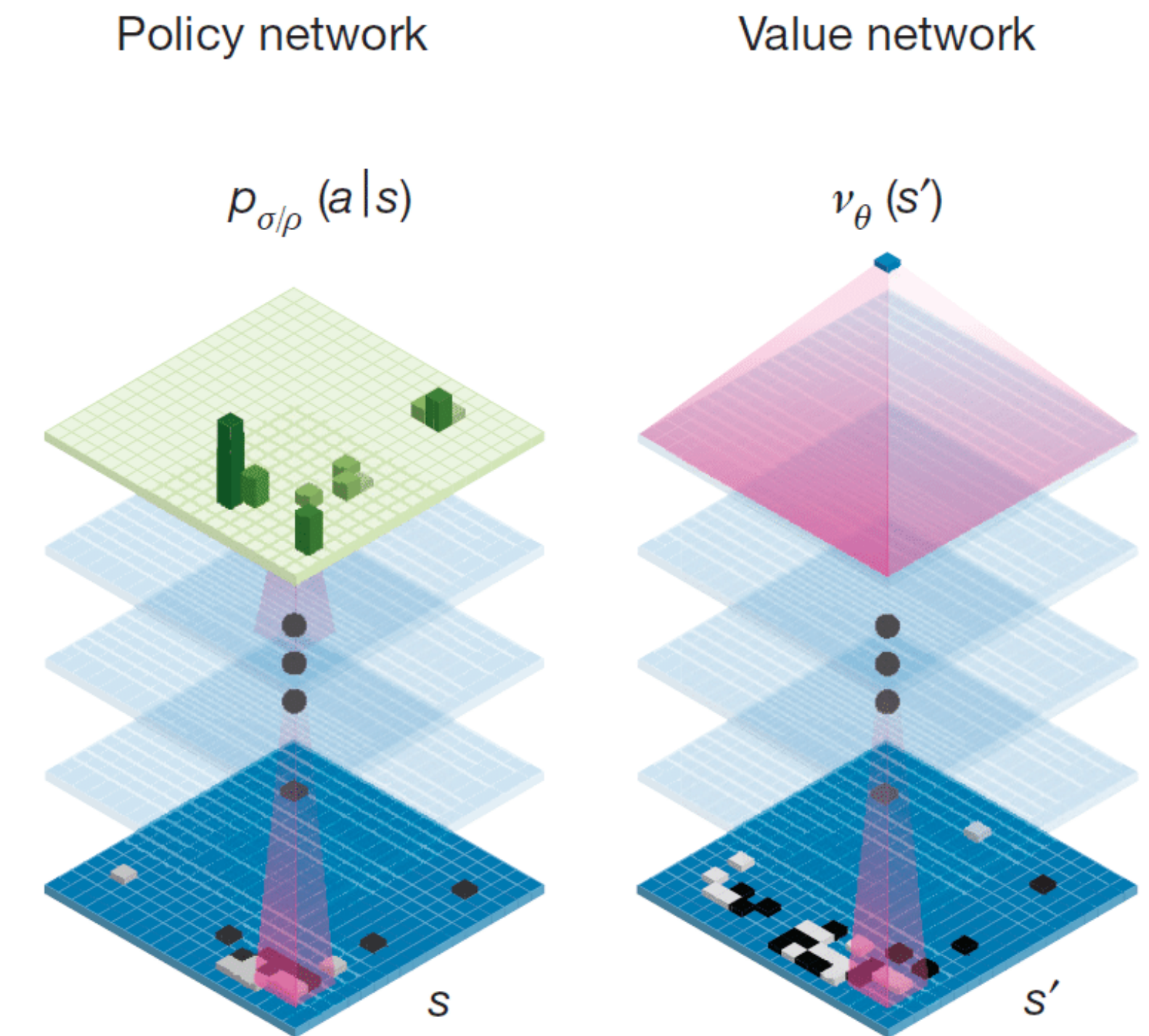
Key characteristics

AlphaGo Zero combines:

- A novel self-play RL algorithm
- Neural network (NN) for function estimation that, given a board's state:
 - Predicts move selection → policy network
 - Predict the game winner → value network
- Monte Carlo Tree Search (MCTS)

Neural Network

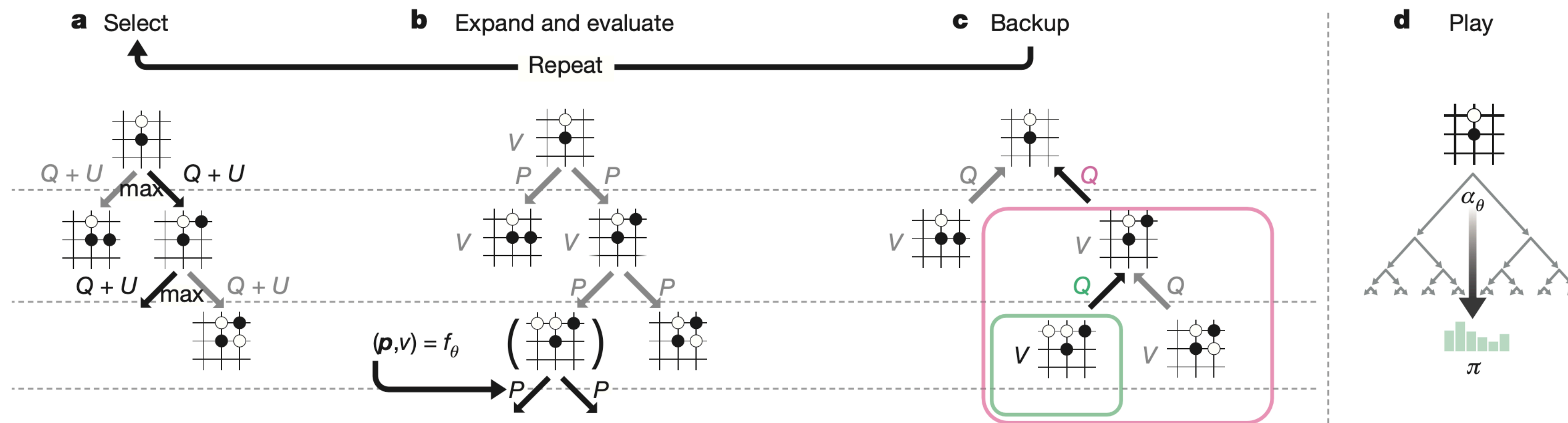
- Residual Neural Network f_θ
- **Input:** board representation s of the
 - current board position (black/white stones)
 - its history (necessary because of forbidden repetitions) $\rightarrow$ Go is not fully observable solely from the current position
- **Output:** move probabilities and value $(p, v) = f_\theta(s)$
 - p : vector of move probabilities, each probability of selecting the move $p_a = \Pr(a | s)$
 - v : scalar, probability of winning of the current player from position s



Credit: Silver et al. 2016

Monte Carlo Tree Search

- Objective: given a state s , output the probabilities π of playing each move
- Simulates games using the current NN f_θ
- For each edge (s, a) store:
 - Prior probability $P(s, a)$
 - Visit count $N(s, a)$
 - Action value $Q(s, a)$



Monte Carlo Tree Search

- **Selection:**

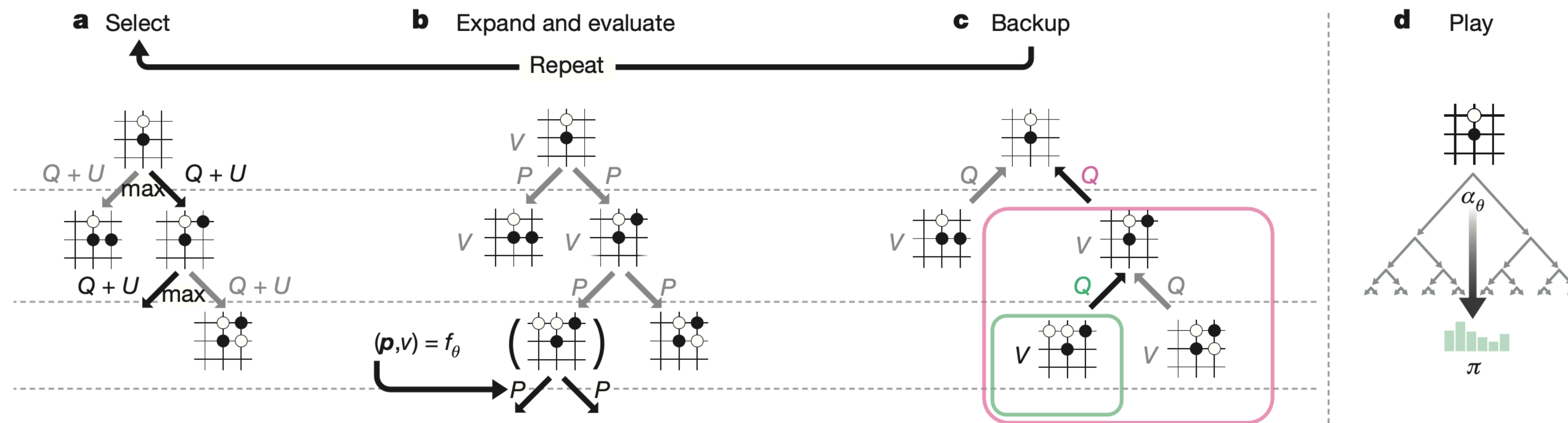
- Traverses the game tree by selecting moves that maximize an upper confidence bound $Q(s, a) + U(s, a)$
- Where $U(s, a) \propto P(s, a)/(1 + N(s, a))$

- **Expansion, Evaluation and Backup:**

- Upon reaching a leaf, add all legal moves to the node; use the network to evaluate a random rotation or reflection of the state
- Propagate the obtained value up the tree, updating Q-values and visit counts for each node along the path

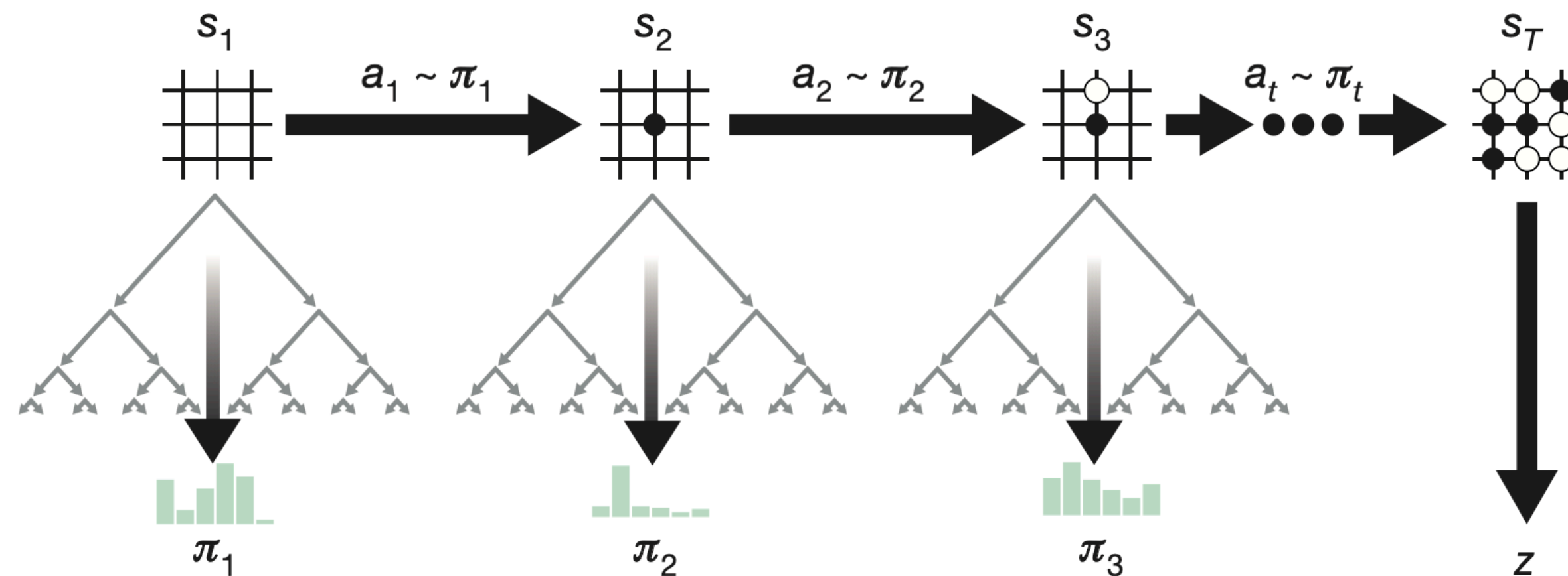
- **Play:**

- Once the search is complete, return the search probabilities
- Sample move according to π



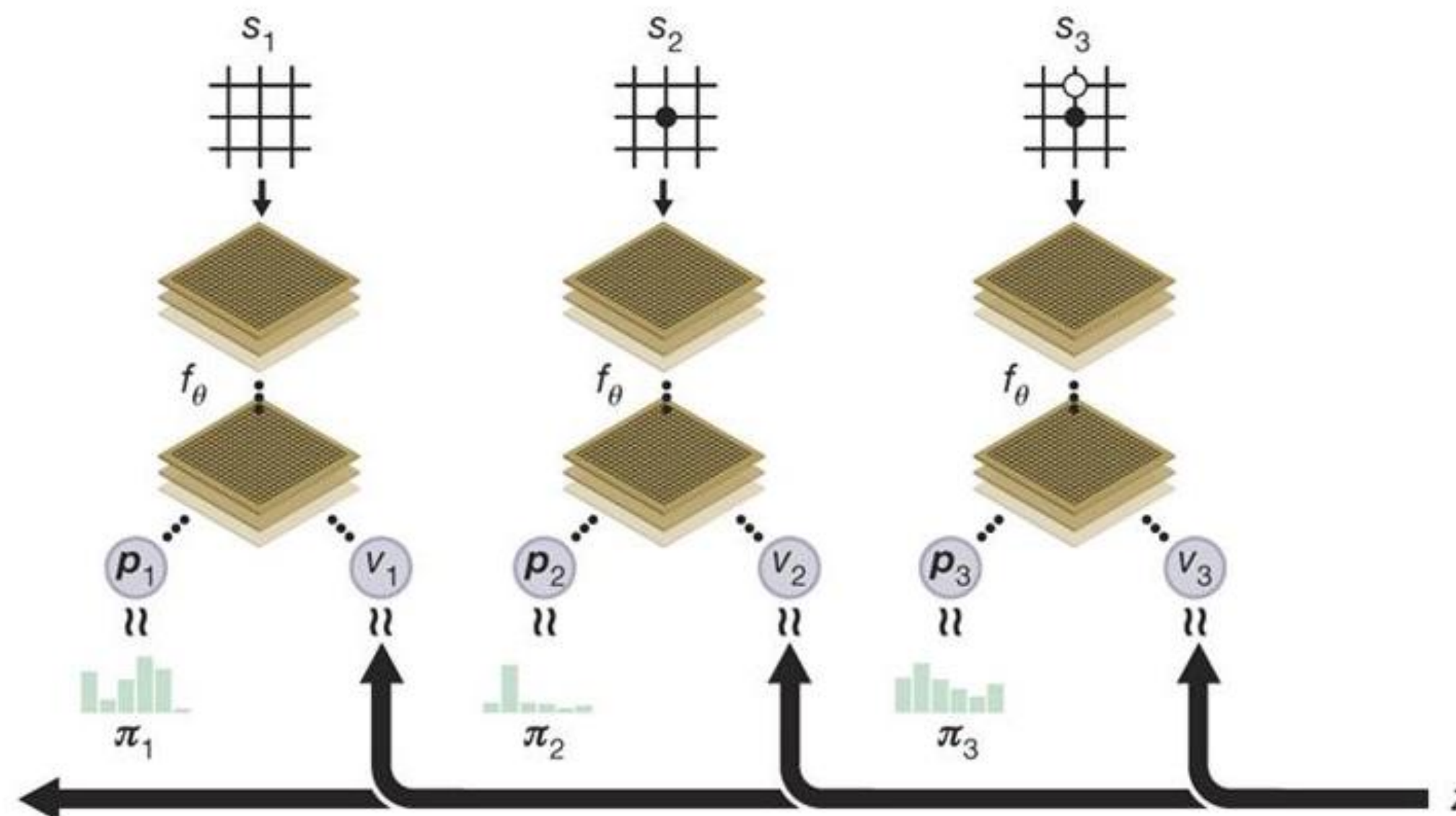
Self-play

- The agent plays full games against itself to generate training data
- At each turn:
 - predict move probabilities and state value for the current state s_t
 - use predicted probabilities to guide MCTS and obtain an improved policy π_t
 - select a move according to the search probabilities
- Terminal state is scored following the game's rules to compute the winner z



Self-play RL algorithm

- During self-play, store the data generated at each step t as a triple (s_t, π_t, z_t)
- Dataset is augmented to include rotations and reflections of each position
- Sample data uniformly among all time-steps of the last games
- Optimize to:
 - Minimize error between predicted value v_t and game winner z_t
 - Maximize similarity between network probabilities p_t and search probabilities π_t



Self-training Pipeline

AlphaGo Zero's self-play training pipeline consists of three main components, all executed asynchronously in parallel:

1. Optimization:

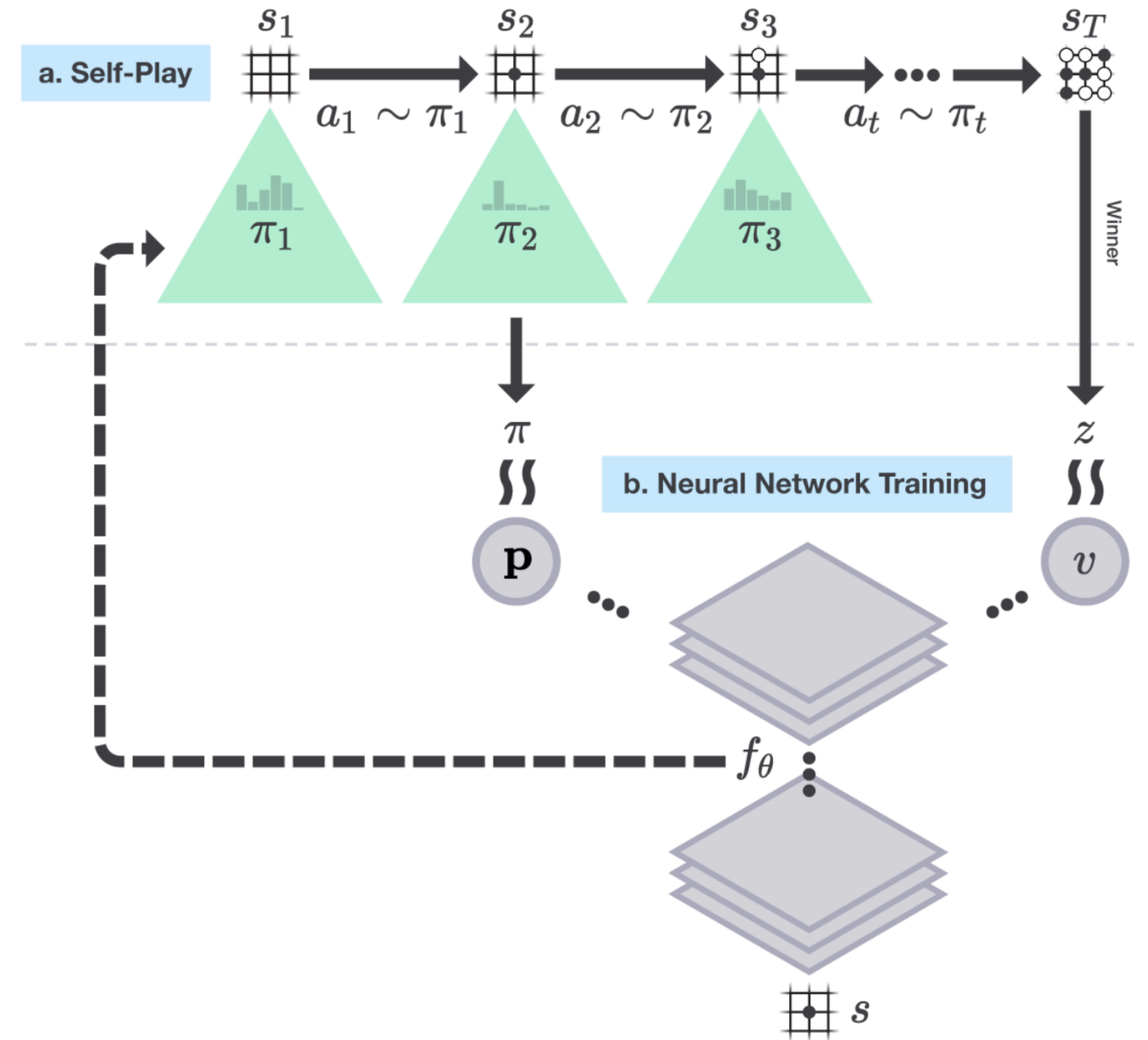
Neural network parameters θ_i are continually optimized from recent self-play data

2. Evaluation:

AlphaGo Zero players α_{θ_i} are continually evaluated

3. Self-play:

The best performing player so far, α_{θ_*} , is used to generate new self-play data



Self-training Pipeline: Optimization

Training Worker

$$l = (z - v)^2 - \boldsymbol{\pi}^T \log \boldsymbol{p} + c \|\boldsymbol{\theta}\|^2$$

1. Optimization

- Train multiple networks in parallel using **stochastic gradient descent** with momentum and learning rate annealing
- Training data (s, π, z) sampled uniformly from the last 500,000 **self-play games**
- Produce a new **checkpoint** every 1000 training steps

Self-training Pipeline: Evaluation

Evaluator

$$\pi' > \pi$$

2. Evaluation

- Each neural network checkpoint is evaluated against the current best f_{θ_*} in 400 games
- If the new player wins by a margin $> 55\%$, it becomes the new best player in the next self-play games

Self-training Pipeline: Self-play

Self-play Worker



The game state

S

π

The search probabilities

π



The winner

Z

3. Self-play

- The current best player α_{θ_*} determined by the evaluator is used to generate data
- In each iteration, α_{θ_*} plays 25,000 games of self-play using 1,600 simulations of MCTS to select each move

Results: Overview

AlphaGo Zero model was tested and evaluated...

... first using:

- 20 block resNet
- 3 days of training
- A single machine in the Google Cloud with 4 TPUs

... then using (final model):

- 40 block resNet
- 40 days of training
- A single machine in the Google Cloud with 4 TPUs

... against:

	Results vs Human Opponents	Description
AlphaGo Fan	5 - 0	Original AlphaGo implementation. Defeated Fan Hui.
AlphaGo Lee	4 - 1	Improved version of AlphaGo. Defeated Lee Sedol.
AlphaGo Master	60 - 0	Same architecture as AlphaGo Zero, trained on human data. Defeated strongest human professionals in online games.

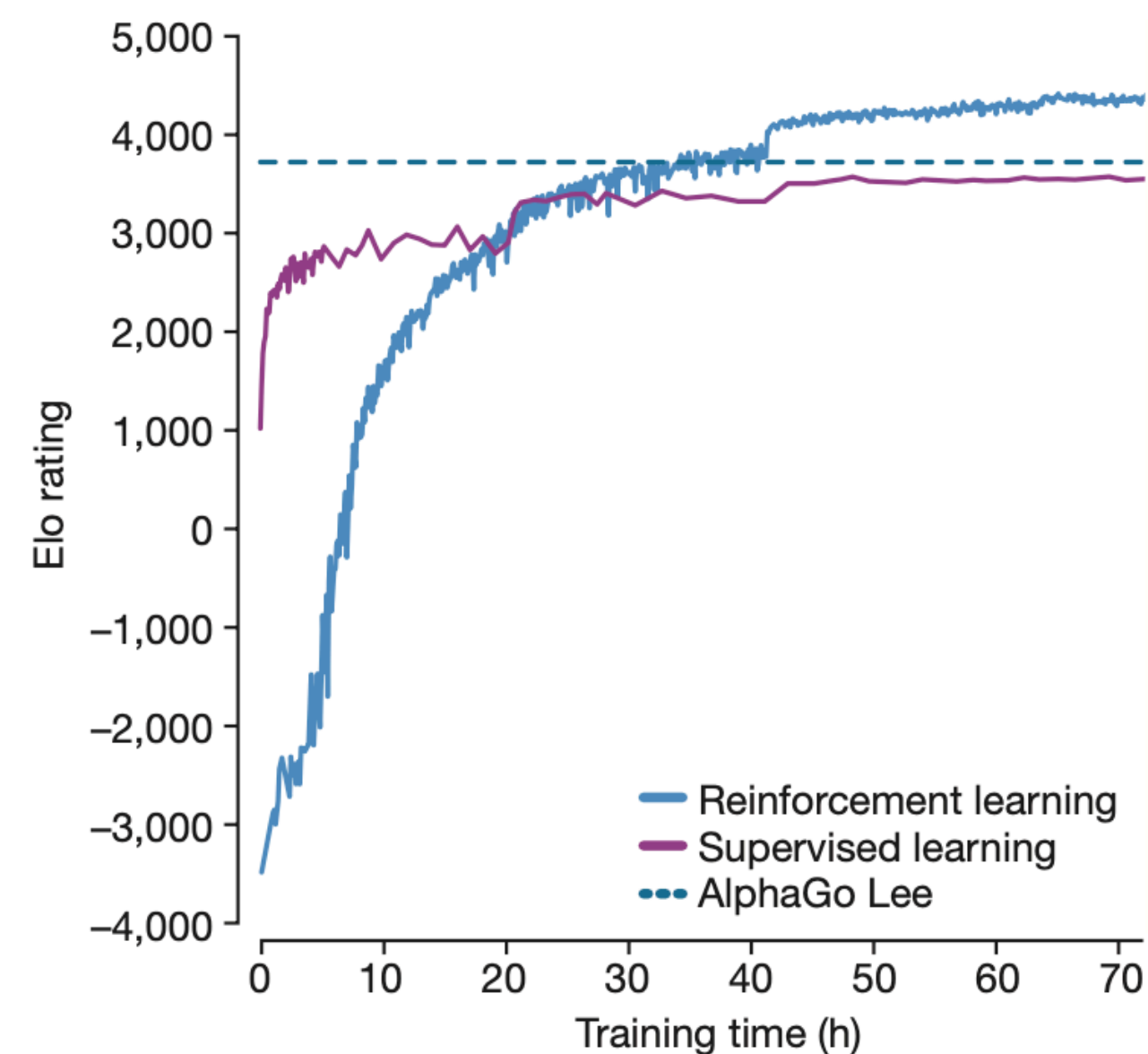
and several previous Go programs like GnuGo, Pachi and Crazy Stone

Results: Empirical Evaluation

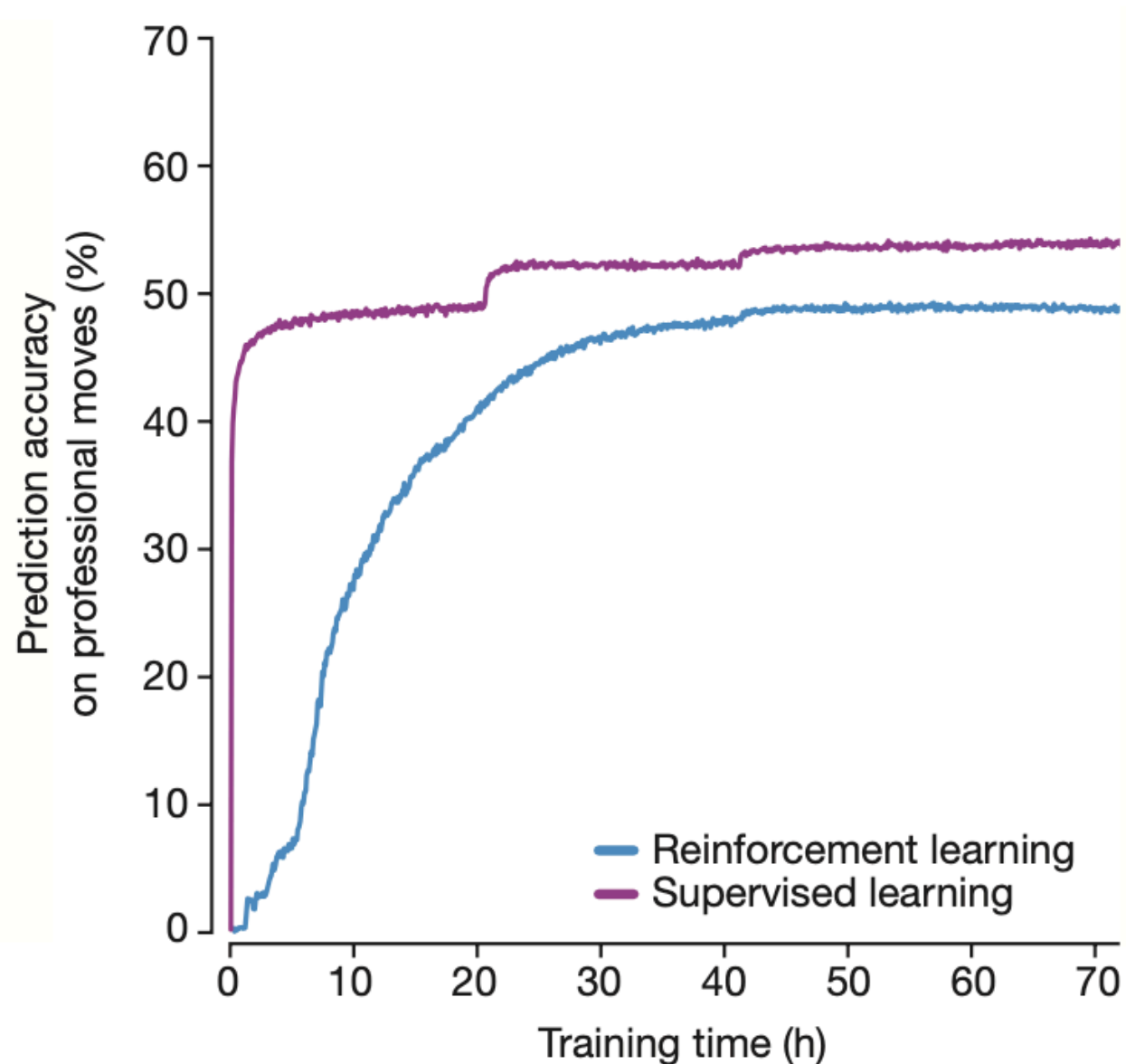
vs. AlphaGo Lee:

- Trained over several months
- Distributed over many machines and used 48 TPUs
- Defeated by 100 games to 0

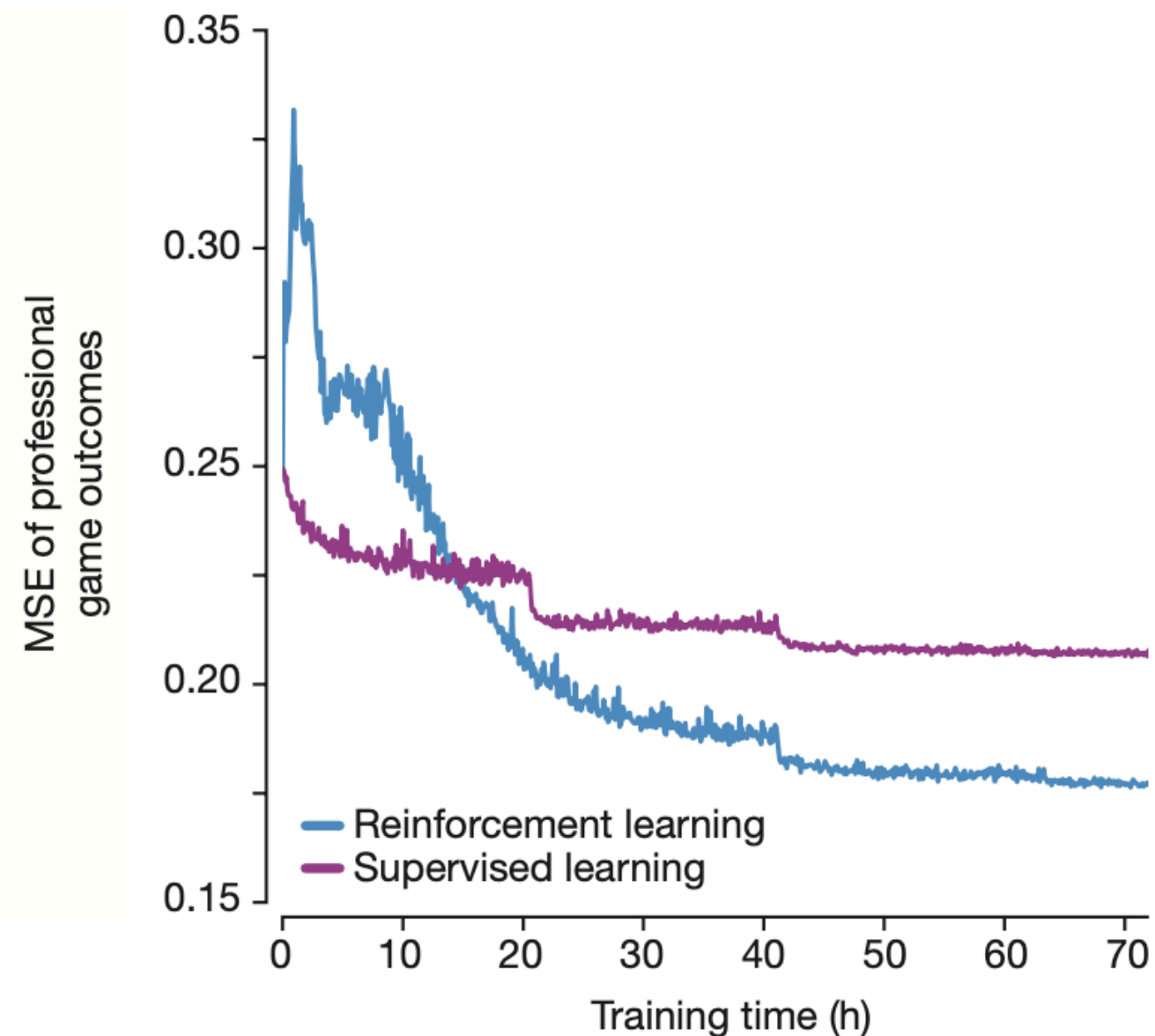
a Performance in the game



b Prediction accuracy on human moves

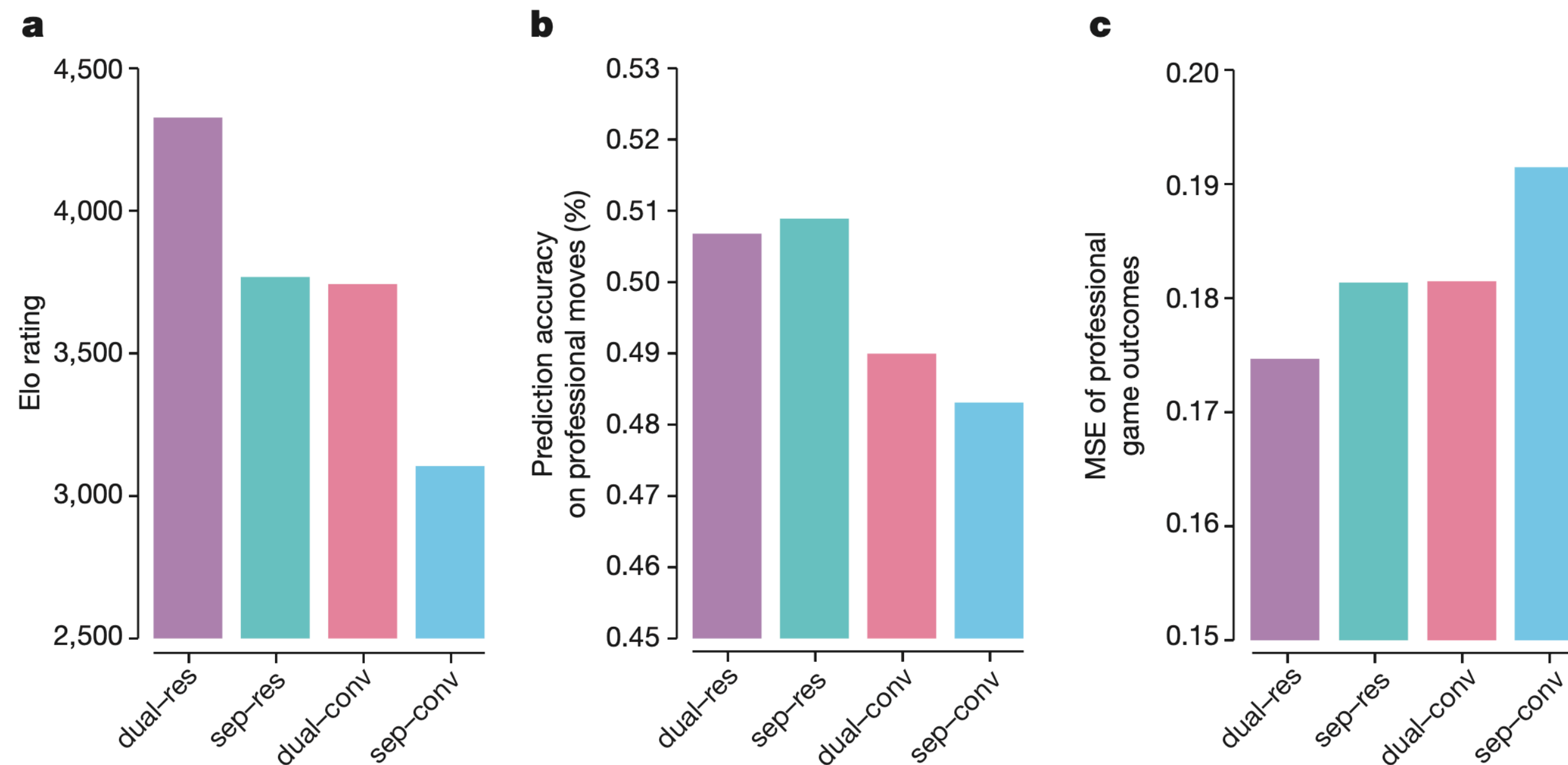


c Game outcomes prediction error



Results: Architectures

- Dual = single NN for both policy and value
- Sep = separate policy and value networks
- Res = residual network
- Conv = convolutional network



Comparison of NN architectures in AlphaGo Zero and AlphaGo Lee

Results: Final performance

Final AlphaGo Zero model evaluated using an internal tournament:

vs. Raw network, AlphaGo Master, AlphaGo Lee, AlphaGo Fan, Crazy Stone, Pachi, GnuGo

Raw network:

- Player based solely on the raw neural network of AlphaGo Zero (without MCTS)
- It simply selects the move with maximum probability

AlphaGo Master:

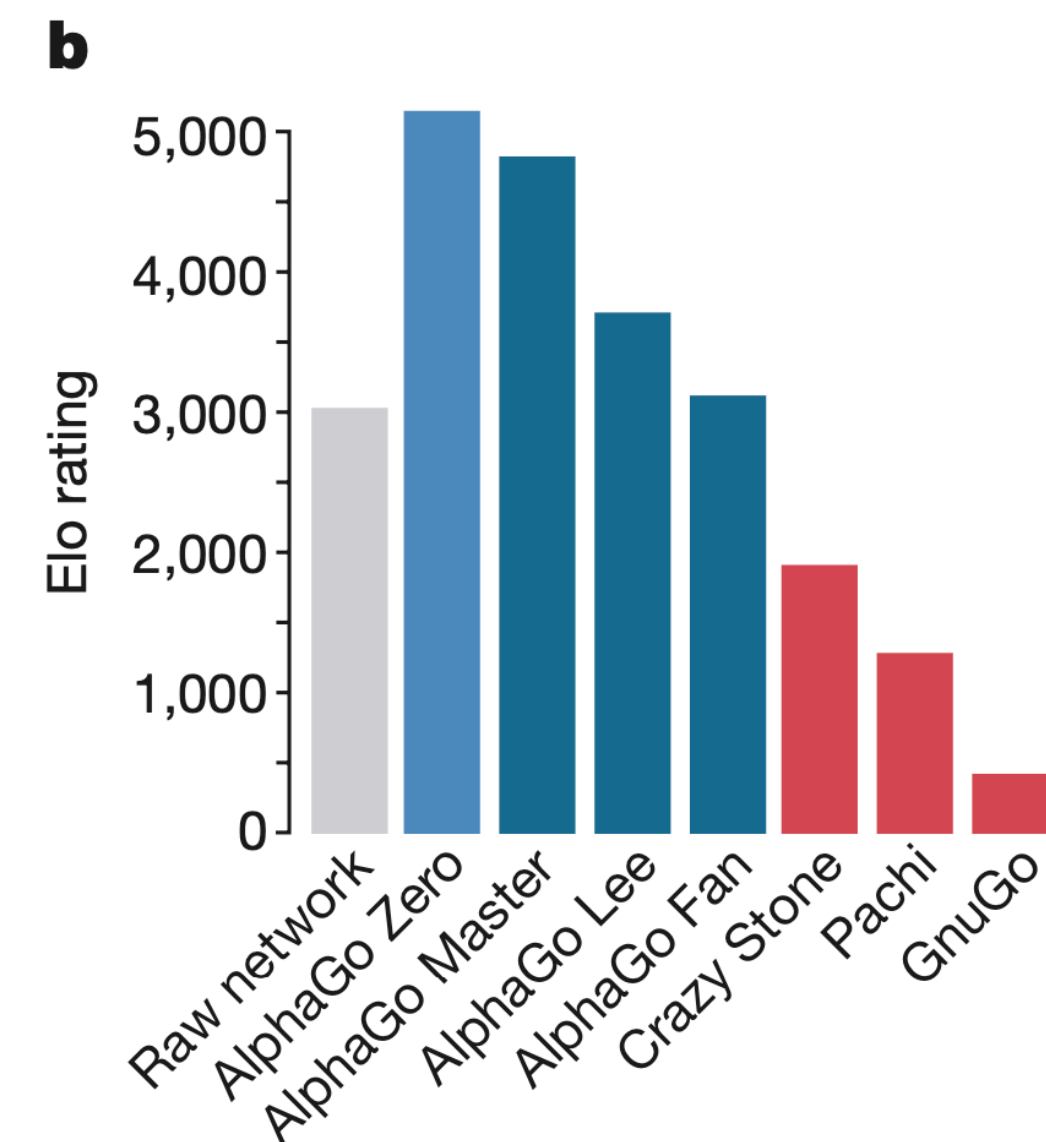
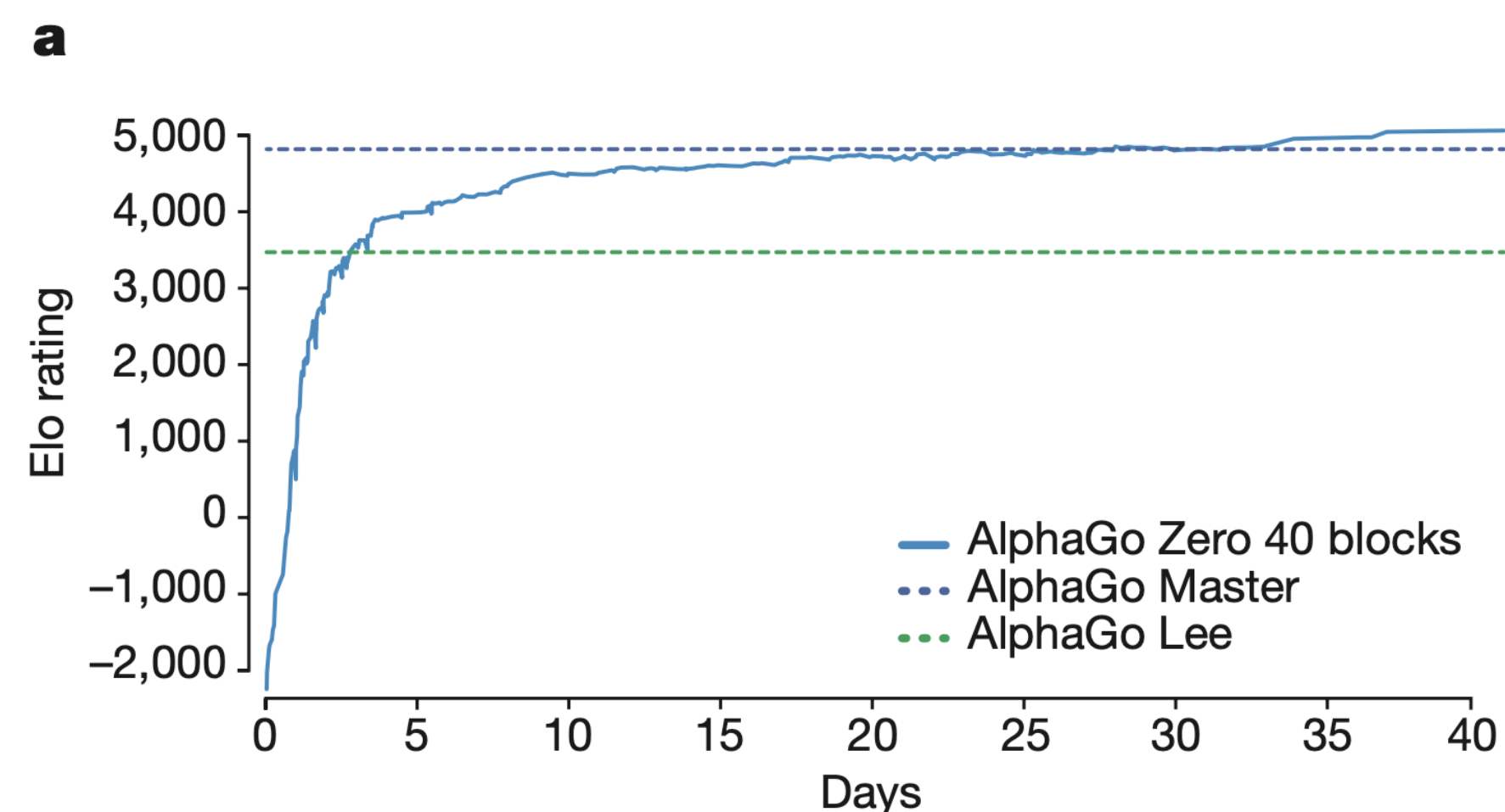
- Distributed on a single machine with 4 TPUs

AlphaGo Lee:

- Distributed over many machines and used 48 TPUs

AlphaGo Fan:

- Distributed over many machines and used 176 TPUs



Performance of AlphaGo Zero

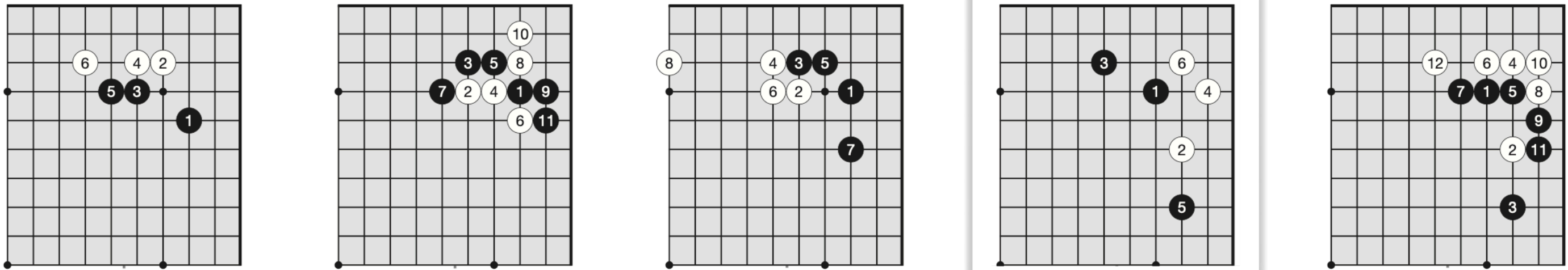
Go knowledge learned by AlphaGo Zero

- The idea of AlphaGo is to discover what it means for a program to be able to learn for itself what knowledge is
- AlphaGo Zero discovered a good level of Go knowledge during its self-play training process:
 - Common patterns and opening of human Go knowledge
 - Non-standard strategies beyond traditional Go knowledge
- Learned a strategy qualitatively different from human play
- Developed **new pieces of knowledge** which were creative and novel in many ways, providing **new insights** into the game of Go

AlphaGo Zero: Discovering new knowledge

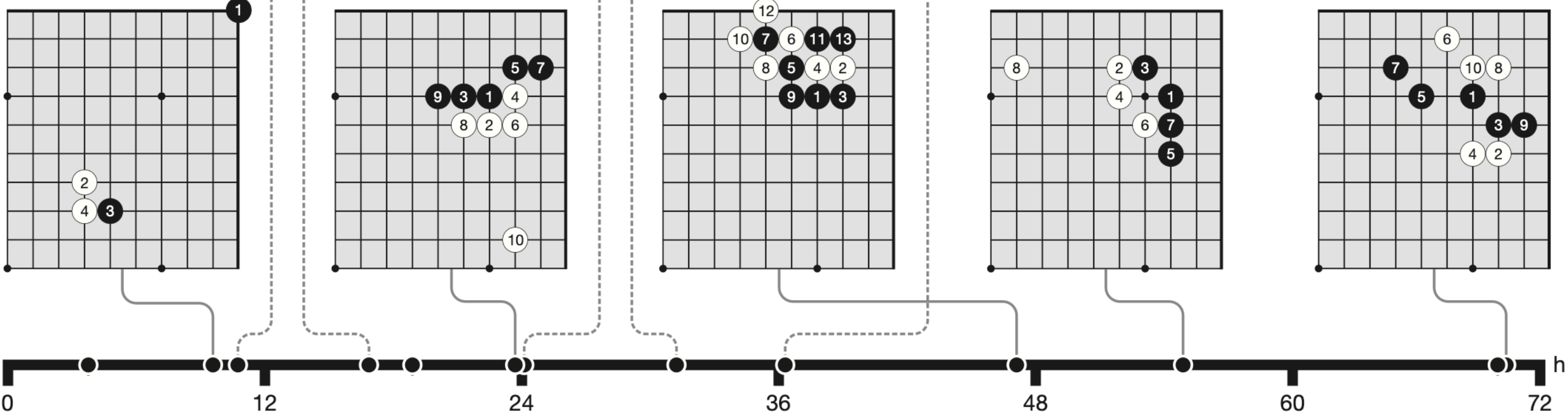
a.

Five human *joseki* discovered during training

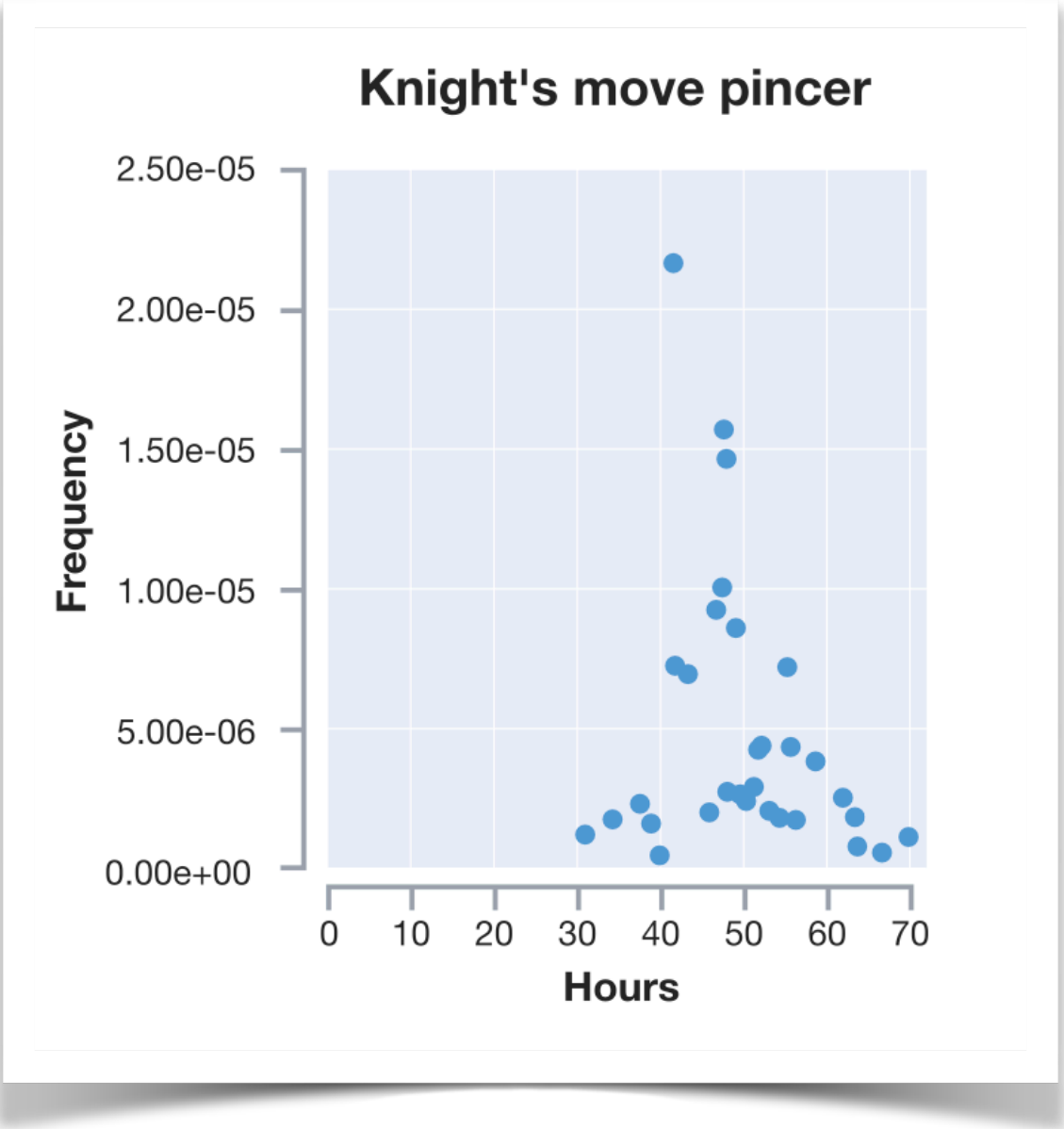


b.

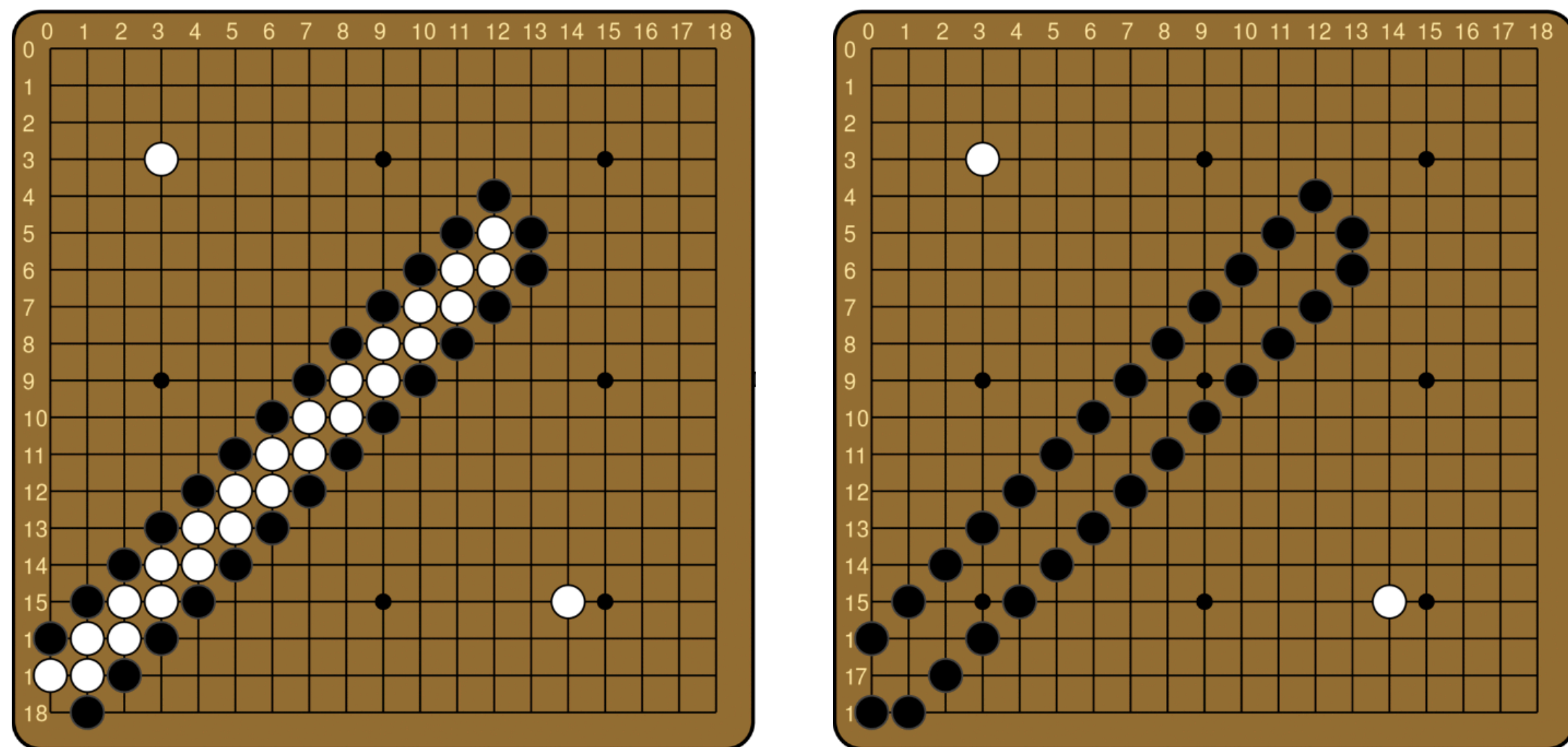
Five *joseki* favoured at different stages of self-play training



joseki = common corner sequences



AlphaGo Zero: Discovering new knowledge



- One of the first elements of Go knowledge learned by humans
- Only understood by AlphaGo Zero much later in the training
- AlphaGo Zero discovered something that was obvious to humans much later, but it also discovered strategies that had never been obvious to human players at all

shicho (japanese for “ladder”) = capture sequence that may span the whole board

Advantages of AlphaGo Zero

- **Simple architecture:** single network for both policy and value, and simple MCTS algorithm
- **Efficiency:** no reliance on rollouts to evaluate states, saves computational resources during search
- **Self-improvement:** no dependency on human data, through self-play reinforcement, it learns
 - starting from random initial weights
 - using only the raw board state and history as input features
 - using the minimum amount of domain knowledge
- **Domain agnostic:** the employed method can be extended to other two-players games with little adjustments

Conclusions

- AlphaGo Zero highlights the potential of **reinforcement learning** in complex decision-making tasks
- It is possible to train to superhuman level, with **no human guidance** and given **no knowledge** of the domain beyond basic game rules
- AlphaGo Zero is able to **overcome** in performance both supervised learning systems and humans
 - Compared to training on human expert data, achieves ***better asymptotic performance*** requiring only a few more hours to train
 - Through self-play and starting tabula rasa, achieves ***superhuman performance*** in the span of a few days of training
 - with **less computational resources** than previous methods
 - rediscovering much of human Go knowledge as well as **novel strategies**

References

Silver, D., Schrittwieser, J., Simonyan, K., Antonoglou, I., Huang, A., Guez, A., ... & Hassabis, D. (2017). *Mastering the game of Go without human knowledge*. Nature, 550(7676), 354–359. <https://doi.org/10.1038/nature24270>

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). MIT Press.

DeepMind. (2017, January 19). AlphaGo – The Movie | Full Documentary. YouTube. <https://www.youtube.com/watch?v=WXuK6gekU1Y>

Thanks for your attention

